

Travel Demand and Shared-Ride Experience in New York City

Data and scope

Analyze the April and May 2026 NYC Taxi and Limousine Commission (TLC) trip records to examine travel demand, passenger spending, sharing patterns, and shared-ride experience by completing all below five tasks using these two datasets:

- Yellow taxi trip records.
- High-volume for-hire vehicle (HVFHV) trip records.

Download only what you need from the NYC TLC Trip Record Data website:

<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

You will need:

- April and May yellow taxi trip records (Parquet file).
- April and May HVFHV trip records (Parquet file).
- The corresponding data dictionaries.
- The taxi-zone lookup table.
- The taxi-zone boundaries (shapefile).

Data Dictionaries and MetaData

- Trip Record User Guide
- Yellow Trips Data Dictionary
- Green Trips Data Dictionary
- FHV Trips Data Dictionary
- High Volume FHV Trips Data Dictionary
- Working With PARQUET Format

Taxi Zone Maps and Lookup Tables

- Taxi Zone Lookup Table (CSV)
- Taxi Zone Shapefile (PARQUET)
- Taxi Zone Map – Bronx (JPG)
- Taxi Zone Map – Brooklyn (JPG)
- Taxi Zone Map – Manhattan (JPG)
- Taxi Zone Map – Queens (JPG)
- Taxi Zone Map – Staten Island (JPG)

Trip Record Data Download Links

Trip data is published monthly on this website, typically with a two-month delay to allow time for full vendor submissions. Due to the size of the datasets, the trip record files have been stored in the PARQUET format. Please see the 'Working With PARQUET Format' under the Data Dictionaries and MetaData section for details. Please be advised that there may be minor changes in the near future to standardize the parquet schema across all years and datasets. If you would like to view the data on NYC Open Data, and export the data in various other formats, you may view the collection [here](#).

Expand All
Collapse All

▼ 2026

January

- Yellow Taxi Trip Records (PARQUET)
- Green Taxi Trip Records (PARQUET)
- For-Hire Vehicle Trip Records (PARQUET)
- High Volume For-Hire Vehicle Trip Records (PARQUET)

February

- Yellow Taxi Trip Records (PARQUET)
- Green Taxi Trip Records (PARQUET)
- For-Hire Vehicle Trip Records (PARQUET)
- High Volume For-Hire Vehicle Trip Records (PARQUET)

March

- Yellow Taxi Trip Records (PARQUET)
- Green Taxi Trip Records (PARQUET)
- For-Hire Vehicle Trip Records (PARQUET)
- High Volume For-Hire Vehicle Trip Records (PARQUET)

April

- Yellow Taxi Trip Records (PARQUET)
- Green Taxi Trip Records (PARQUET)
- For-Hire Vehicle Trip Records (PARQUET)
- High Volume For-Hire Vehicle Trip Records (PARQUET)

May

- Yellow Taxi Trip Records (PARQUET)
- Green Taxi Trip Records (PARQUET)
- High Volume For-Hire Vehicle Trip Records (PARQUET)

When completing the tasks for this assignment, apply the following conventions throughout:

- For some questions, you need to identify Uber and Lyft using *hvfhs_license_num*.
- Do not assume that all recorded values are correct.
- Assign trips to month, day, and hour using the pickup timestamp, not the drop-off or request timestamp.
- Treat origin–destination (OD) pairs as directed: A → B differs from B → A.
- Retain within-zone trips, i.e., including A → A.
- Interpret counts as recorded trips, not unique people or vehicles.
- Interpret demand as observed completed trips, rather than all travel requests.
- Yellow taxi sharing status is unspecified. Use HVFHV data only for analyses of sharing.

Task 1: Prepare audit and summarize the data [15%]

Develop a validation process to identify missing, invalid, and inconsistent records. Review the data dictionaries to understand what each field represents and which fields each analysis requires.

1.1: Clean and audit the data

Check for the following issues:

- Missing or invalid zone IDs.
- Missing timestamps, nonpositive trip durations, and inconsistent timestamp sequences.
- Records outside the study period.
- Missing or implausible distances and payments.
- Missing or inconsistent HVFHV sharing flags.

This data-cleaning step should be implemented within the relevant subsequent analyses. Apply each cleaning rule only when the corresponding data issue affects that analysis.

- State and justify the rule.
- Report the number of records affected.
- Report any overlap between rules.
- Report the number of records retained for each analysis.

1.2: Count trips by category

Complete the table below separately for April and May. Use share request flag first and match flag second (for example, Y (Yes) / N (No) means a sharing request without a reported match). Report any other company codes separately by adding rows at the bottom of this table so that company totals reconcile with the HVFHV total.

Service	Request / match category	April trips	May trips
Yellow taxi	All retained trips		
HVFHV	All retained trips		
Uber	N / N		
Uber	Y / N		
Uber	Y / Y		
Uber	N / Y		
Uber	Missing or other statuses		
Lyft	N / N		
Lyft	Y / N		
Lyft	Y / Y		
Lyft	N / Y		
Lyft	Missing or other statuses		
...			

Task 2: Identify spatial patterns of demand and spending [25%]

Conduct each analysis separately for April and May in 2026. Compare services within each month and describe changes between months.

2.1: Map origin and destination hotspots

Produce four maps for each month (eight maps for April and May in total), showing:

- Yellow taxi trips originating in each zone.
- Yellow taxi trips ending in each zone.
- HVFHV trips originating in each zone.
- HVFHV trips ending in each zone.

Identify the leading zones and explain the differences across maps. Use comparable legends and clearly label the month, service, and measure.

You may use Python/GeoPandas, QGIS, ArcGIS, or another tool to render the maps.

2.2: Identify the largest OD flows of trip volumes

For yellow taxis, Uber, and Lyft separately, identify the top ten directed OD flows in terms of their trip volumes in each month. For each of the top 10 flows, report its origin and destination zones, trip count, and its share of the relevant service's total trips for that month. State the denominator value.

Map the leading OD flows and use the maps to Assess whether the dominant flows are similar across yellow taxis, Uber, and Lyft.

2.3: Identify the largest OD flows in terms of passenger spendings

For each month and service (yellow taxis, Uber, and Lyft), identify the top ten directed OD flows with the greatest total passenger spending. For each of the top 10 flows, report its total number of trips with valid spending values and their mean payment to distinguish high volume from high spending per trip.

- Yellow taxi: use total_amount.
- HVFHV: sum base_passenger_fare, tolls, bcf, sales_tax, congestion_surcharge, airport_fee, tips, and cbd_congestion_fee.

Do not add driver_pay to passenger spending or add fare components to total_amount again. Explain how missing components are handled; do not silently treat missing values as zero. Yellow taxi totals exclude unrecorded cash tips, so the measure does not capture all money paid.

Map the top OD flows by passenger spending and explain how their patterns differ from those based on trip counts.

Task 3: Measure sharing patterns and reporting consistency [15%]

Use HVFHV data only. Analyze Uber and Lyft separately for each month.

Use `shared_request_flag` to identify whether a shared ride was requested and `shared_match_flag` to identify whether the trip found a match.

3.1 Define the measures

Let n_{NN} , n_{NY} , n_{YN} , and n_{YY} denote trip counts indexed by request flag first and match flag second, corresponding to the "Request/Match Category" column in Task 1.2.

Define V as their sum:

$$V = n_{NN} + n_{NY} + n_{YN} + n_{YY}$$

$$\text{Pooling request rate} = \frac{n_{YN} + n_{YY}}{V}$$

$$\text{Reported matched rate} = \frac{n_{NY} + n_{YY}}{V}$$

$$\text{Matching success among requests} = \frac{n_{YY}}{n_{YN} + n_{YY}}$$

Exclude missing or other flag statuses from these denominators and report them separately. Treat a rate with a zero denominator as undefined, not zero. Multiply a rate by 100 when reporting it as a percentage.

3.2 Locate sharing concentrations

By pickup zone and month, map the reported matched rate and matching success among requests.

For comparisons and rankings across zones, include only zones that meet the minimum sample-size requirements below:

- A zone must have at least 100 valid-status trips to report the reported matched rate,
- Or at least 100 pooling requests to report matching success among requests.

Always report the relevant denominator. Zones that do not meet the threshold should be labeled as “insufficient sample” rather than a value of zero

Produce maps to show the spatial distribution of **pooling request rates** and **matching success rates**, by Uber and Lyft, respectively.

Task 4: Estimate relative excess travel time [25%]

For this analysis, use Uber data only. The purpose of this task is to analyze whether shared trips took significantly longer time than non-shared trips for the same pair of origin and destination zones.

4.1: Construct a non-shared reference

Use Uber trips with request N / match N as the **non-shared reference** data. Define each reference cell using five dimensions below:

origin zone × destination zone × pickup hour × weekday/weekend;

where weekday represents Monday–Friday and weekend represents Saturday–Sunday.

Express durations in minutes (if using trip_time, convert seconds to minutes). Pool trips when they fall into the same cell. For example, weekday trips (non-shared) from Zone A to Zone B departing between 08:00 and 08:59 belong to the same reference cell.

Calculate the median trip duration and sample size for each reference cell. Only cells with at least 30 reference trips should be used for comparison. This threshold is specified for this assignment and is not a universal statistical standard.

4.2 Compare shared trips with non-shared reference

Use Uber trips with request Y / match Y as the **shared-trip** data. For each shared trip i , retrieve the non-shared reference defined in 4.1 for its cell, and calculate:

$$E_i = T_i - T_{\text{ref},i}$$

$$R_i = 100 \times \frac{T_i - T_{\text{ref},i}}{T_{\text{ref},i}}$$

Use the following definitions, with both observed and reference durations expressed in minutes:

- T_i is the observed shared-trip duration;
- $T_{\text{ref},i}$ is the median duration of its non-shared reference cell defined in 4.1.
- E_i is excess travel time in minutes.
- R_i is relative excess travel time, expressed as a percentage.

Keep negative values. Do not force differences to be positive.

For trips without a supported reference, $E_i = \text{null}$. Report the percentage of shared trips that have a reference cell to compare with.

Map the average of $\{E_{zi} \mid E_{zi} \text{ is not null}\}$ for each pickup zone Z , where its total number of eligible shared trips is at least 10.

4.3 Summarize and present the comparison results

Summarize the travel-time comparison results as follows:

- Median and interquartile range of excess minutes and relative excess time.
- Differences by pickup hour and month.
- Geographic patterns by pickup zone.
- Results for within-zone versus between-zone trips.

Calculate relative differences for individual trips before summarizing them. State whether summaries weight trips equally or weight OD cells equally.

4.4 Evaluate the interpretation

Explain why this measures an observed difference, not necessarily the causal effect of pooling.

Discuss:

- Different exact endpoints within the same zones.
- Traffic variation across dates within a reference cell.
- Route choice and passenger selection.
- The absence of verified sharing partners and overlap times.
- Unequal reference coverage across locations.

Note:

1. The reference median is not an optimal or congestion-free travel time.
2. The actual trip distance may be longer than the optimal route because of pooling.

Required output: A reference-table description, coverage statistics, travel-time comparison figures, and a critical interpretation.

Task 5: Complete one of the following two tasks [20%]. If you can complete both tasks, you get additional 10% bonus points.

These two tasks are open tasks that you need to design your analysis steps, evaluate results, and draw your conclusions.

5.1: Are shared trips significantly cheaper than non-shared trips?

5.2: How many of the non-shared trips are potentially sharable (in other words, they may find a match if they request sharing)?

Submission requirements

Submit:

- No specific report format or template is required. Students may structure the report as appropriate, provided that the analyses, results, and discussion for all five tasks are presented clearly and concisely.
- Page limit: 10 pages including tables, excluding maps and codes.
- Reproducible code: a notebook or scripts with clear execution instructions and package requirements.
- Supplement files: CSV files supporting the main results.

Do not submit the raw monthly files.

Assessment criteria:

Assessment will emphasize correct definitions and processing, reproducibility, clear visual communication, and sound conclusions supported by the analysis results.